Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 2.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Given that c and d are non-zero integers, the expression $10^{c-2d} \times \frac{20^{2c+d}}{8^c \cdot 125^{c+d}}$ is an integer if

A) $c < 0$ B) $d < 0$ C) $c < 0$ and $d < 0$ D) $c < 0$ and $d > 0$
E) $c > 0$ and $d < 0$ F) $c > 0$ and $d > 0$ G) $d > 0$ H) $c > 0$

2. If $a = b^3$, and $b = \frac{\sqrt{5}}{c}$, what is the value of a when $c = \frac{1}{3}$?

(A) 0.42 (B) 1.89 (C) 60.37 (D) 100.62 (E) 301.87

3. For all $def \neq 0$,

$$\frac{4d^3e^4f^2}{(2de^3f)^2} =$$

(A) $2de^2$ (B) $2de^{-2}$ (C) de^2 (D) de^{-2} (E) $\frac{de^2}{2}$

4. If an arc with a length of 12π is $\frac{3}{4}$ of the circumference of a circle, what is the shortest distance between the endpoints of the arc?

(A) 4 (B) $4\sqrt{2}$ (C) 8 (D) $8\sqrt{2}$ (E) 16

5. A cube with edges of length b is divided into 8 equal smaller cubes. What is the difference between the combined surface area of the 8 smaller cubes and the surface area of the original cube?

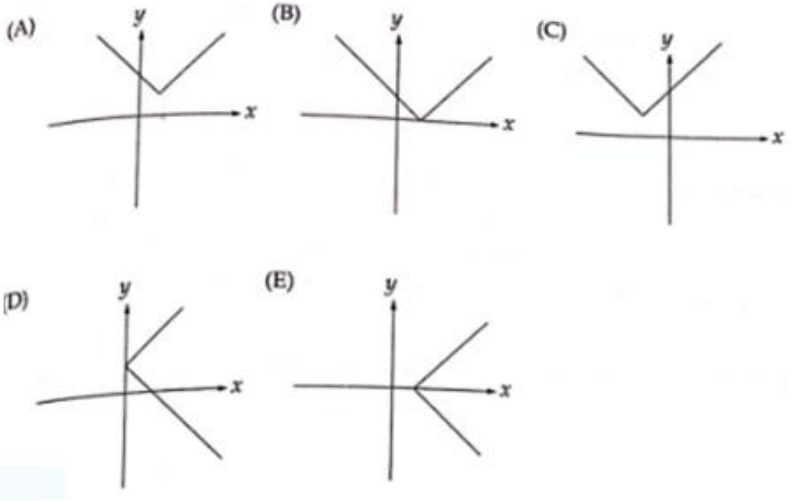
(A) $\frac{3}{2}b^2$ (B) $\frac{3}{4}b^2$ (C) $\frac{9}{2}b^2$ (D) $6b^2$ (E) $12b^2$

6. The ratio of DE to EF is 4 to 9. If $DF = 2$, what is the distance from D to the midpoint of EF?



(A) $\frac{9}{13}$ (B) $\frac{12}{13}$ (C) $\frac{14}{13}$ (D) $\frac{16}{13}$ (E) $\frac{17}{13}$

7. If $p(x) = |2 - x| + \frac{1}{2}$, which of the following could be the graph of $y = p(x)$?



8. If all four interior angles of quadrilateral P have the same measure, which of the following statements must be true?

I. All sides of P have equal length.

II. The diagonals of P are perpendicular.

III. The measure of each interior angle of P is $90°$.

(A) None (B) I only (C) II only (D) III only (E) I, II, and III

9. Let $A = 2014 + \frac{1}{2014}$, $B = 2014 + \frac{1}{2014+\frac{1}{2014}}$, $C = 2014 + \frac{1}{2014+\frac{1}{2014+\frac{1}{2014}}}$. The numbers A, B, and C arranged in increasing order are

(A) A, B, C (B) C, A, B (C) B, C, A (D) C, B, A (E) B, A, C

10. The graph of the equation $x^2 + y^2 = 169$ includes how many points (x, y) in the coordinate plane were x and y are both negative integers?

(A) None (B) One (C) Two (D) Three (E) Infinitely many

11. If $f(x) = x^2 + 2x - 2$ and if $f(s - 1) = 1$, what is the smallest possible value of s?

(A) -3 (B) -2 (C) -1 (D) 1 (E) 2

12. It is given that the expansion of $(ax+b)^3$ is $8x^3 - px^2 + 18x - 3\sqrt{3}$, where a, b and p are real constants.

What is the value of p?

A) $-12\sqrt{3}$ B) $-6\sqrt{3}$ C)$-4\sqrt{3}$ D) $-\sqrt{3}$
E) $\sqrt{3}$ F) $4\sqrt{3}$ G) $6\sqrt{3}$ H) $12\sqrt{3}$

13. What is the sum of the infinite geometric series $2 + \left(-\frac{1}{2}\right) + \left(\frac{1}{8}\right) + \left(-\frac{1}{32}\right) + \cdots$?

(A) $1\frac{3}{8}$ (B) $1\frac{2}{5}$ (C) $1\frac{1}{2}$ (D) $1\frac{3}{5}$ (E) $1\frac{5}{8}$

14. A fair coin is tossed five times. What is the probability that it lands head-up at least twice?

(A) $\frac{3}{16}$ (B) $\frac{1}{4}$ (C) $\frac{7}{16}$ (D) $\frac{13}{16}$ (E) $\frac{27}{32}$

15. Of the 12 members of a high school drama club, 8 are seniors. The club plans to establish an 8-member committee to interview potential club members. If exactly 6 members of the committee must be seniors, how many committees are possible?

(A) 21 (B) 34 (C) 168 (D) 336 (E) 495

16. 60% of a sports club's members are women and the remainders are men. This sport club offers the opportunity to play tennis or cricket. Every number plays exactly one of the two sports.

$\frac{2}{5}$ of the male members of the club play cricket;

$\frac{2}{3}$ of the cricketing members of the club are women.

What is the probability that a member of the club, chosen at random, is a woman who plays tennis?

(A) $\frac{1}{5}$ (B) $\frac{7}{25}$ (C) $\frac{1}{3}$ (D) $\frac{11}{25}$ (E) $\frac{3}{5}$

17. For what value of x is $|16 - (x+5)^2|$ at its minimum?

(A) -9 (B) -5 (C) 5 (D) 9 (E) 16

18. If f(x) = $\frac{1}{\sqrt{1-x^2}}$, which of the following describes all the real values of x for which f(x) is undefined?

(A) $x = -1$ or $x = 1$ (B) $x < -1$ or $x > 1$ (C) $x \leq -1$ or $x \geq 1$ (D) $-1 < x < 1$ (E) $-1 \leq x \leq 1$

19. Kayla drove from Bayside to Chatham at a constant speed of 21 miles per hour and then returned along the same route from Chatham to Bayside. If her average speed for the entire journey was 26.25 miles per hour, at what average speed, in miles per hour, did Kayla return from Chatham to Bayside?

(A) 28 (B) 31 (C) 31.5 (D) 35 (E) It cannot be determined from the information given.

20. The average (arithmetic mean) of six numbers is 3. If the average of the least and the greatest of these numbers is 5, then the other four numbers could be any of the following EXCEPT

(A) $-4, 0, 5, 7$ (B) $0, 0, 0, 8$ (C) $\frac{1}{2}, \frac{3}{2}, 1, 5$ (D) 1, 1, 1, 1 (E) 1, 2, 2, 3

21. Today Anselm is three times as old as his brother Bartholomew, and Bartholomew is 4 years younger than his sister Catheromme. If Anselm, Bartholomew, and Catherine are all alive 5 years from today, which of the following must be true on that day?

I. Anselm is three times as old as Bartholomew.

II. Catherine is 4 years older than Bartholomew.

III. Anselm is older than Catherine

(A) I only (B) II only (C) III only (D) I and II (E) II and III

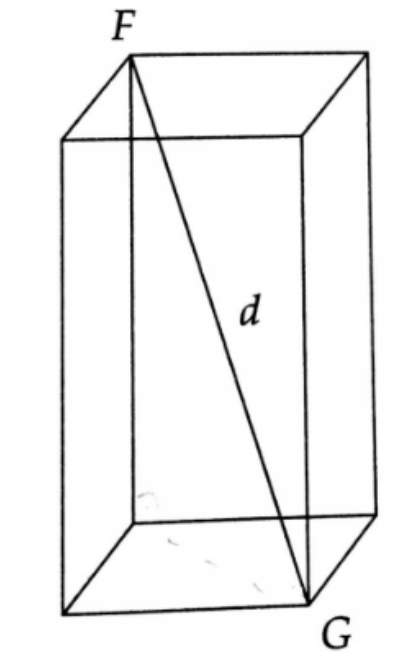
22. Consider the statement:

(*) A whole number n is prime if it is 1 less or 5 less than a multiple of 6.

How many counterexamples to (*) are there in the range $0 < n < 50$?

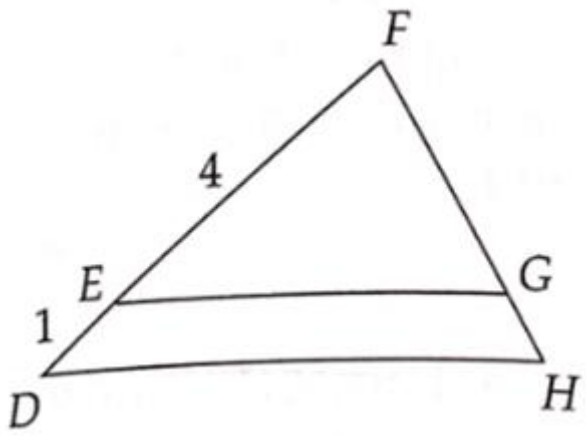
(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

23. In parallelepiped, d is the distance from vertex F to vertex G. The base is square, and the height is twice the width. What is the volume of the solid in terms of d?



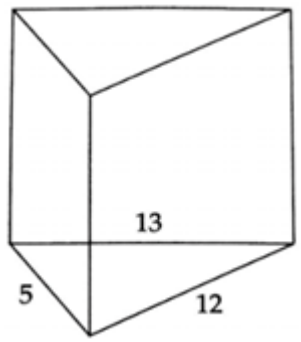
(A) $12d^3\sqrt{6}$ (B) $10d^3\sqrt{5}$ (C) $6d^3\sqrt{2}$ (D) $\frac{2d^3\sqrt{5}}{5}$ (E) $\frac{d^3\sqrt{6}}{18}$

24. $EG \parallel DH$, and the lengths of segments DE and EF are as marked. If the area of EFG is a, what is the area of DFH in terms of a?



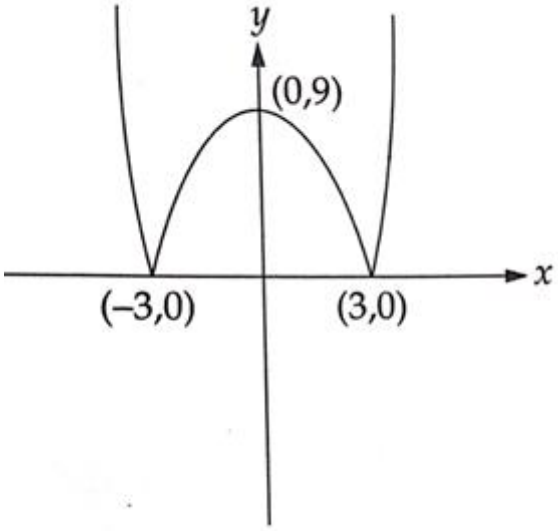
(A) $\frac{4a}{5}$ (B) $\frac{16a}{25}$ (C) $\frac{16a}{20}$ (D) $\frac{5a}{4}$ (E) $\frac{25a}{16}$

25. In the prism, the bases of the right uniform solid are triangles with sides of lengths 5, 12, and 13. If the surface area of the solid is 360, what is the volume?



(A) 30 (B) 180 (C) 300 (D) 600 (E) 780

26. The graph could be the graph of which of the following functions?



(A) $f(x) = x^2 + 9$ (B) $f(x) = (x-9)^2$ (C) $f(x) = 9 - x^2$ (D) $f(x) = -|x^2 + 9|$ (E) $f(x) = |-x^2 + 9|$

27. The "panvoid" of a function is defined as the sum of the integers that do <u>not</u> fall within the domain of the function. All of the following functions have equal panvoids EXCEPT

(A) $f(x) = \frac{3x}{x^2-4}$ (B) $f(x) = \frac{3-x}{x^3-x}$ (C) $f(x) = \frac{3x}{3x^2-27}$ (D) $f(x) = \frac{x+2}{x}$ (E) $f(x) = \frac{2-x}{x^2-x}$

28. A region is defined by the inequalities $x + y > 6$ and $x - y > -4$

Consider the three statements:

1 $x > 1$

2 $y > 5$

3 $(x + y)(x - y) > -24$

Which of the above statements is/are true for **every** point in the region?

A) none B) 1 only C) 2 only D) 3 only
E) 1 and 2 only F) 1 and 3 only G) 2 and 3 only H) 1, 2 and 3

29. If $3s + 5t = 10$, and $2s - t = 7$, what is the value of $\frac{1}{2}s + 3t$?

(A) 0 (B) 1.5 (C) 2.8 (D) 3.4 (E) 3.5

30. If $p = \frac{\sqrt{q}+z}{r^2+z}$, what is the value of z in terms of p, q, and r?

(A) $\frac{p-r^2\sqrt{q}}{p-1}$ (B) $\frac{pr^2-\sqrt{q}}{r^2}$ (C) $\sqrt{\frac{p+qr}{r^2}}$ (D) $\frac{r^2\sqrt{q}-p}{p+1}$ (E) $\frac{\sqrt{q}-pr^2}{p-1}$

Answer key:

№	Correct Answer	Your answer
1	E	
2	E	
3	D	
4	D	
5	D	
6	E	
7	A	
8	E	
9	D	
10	E	
11	B	
12	H	
13	D	
14	D	
15	C	
16	B	
17	A	
18	D	
19	D	
20	D	
21	E	
22	C	
23	E	
24	E	
25	C	
26	E	
27	E	
28	A	
29	B	
30	E	
		Total: _____